

Sanika Govardhan Sangvai

Data Scientist | ML Engineer

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LinkedIn | GitHub | Portfolio | LeetCode

Education

VIT Bhopal University

B.Tech, Computer Science and Engineering (Health Informatics)

2023 – 2027

CGPA: 8.62/10

Skills

- **Languages:** Python, SQL, C++
- **Machine Learning & Deep Learning:** Machine Learning, Deep Learning, TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, SHAP
- **Tools & Deployment:** FastAPI, Streamlit, MySQL, Git, GitHub

Projects

Customer Churn Prediction & Retention Analytics

Sep 2026

Python, Scikit-learn, XGBoost, SHAP, FastAPI, Streamlit

GitHub | Live Demo

- Built an end-to-end churn prediction system (preprocessing, feature engineering, model training/tuning, explainability, deployment) on the IBM Telco dataset (7,043 customers) using leakage-safe Scikit-learn pipelines
- Trained and compared Logistic Regression, Random Forest, and XGBoost via RandomizedSearchCV (5-fold CV), selecting on ROC-AUC and recall for the churn use case; best model reached **0.838 ROC-AUC and 80% recall** on held-out test data
- Built SHAP-based explainability (global and per-customer) and Low/Medium/High risk segmentation, served via a FastAPI inference endpoint and a live Streamlit dashboard for retention analytics

CheapPill.in – Generic Medicine Price Comparison Platform

Sep 2026

Python, SQL, SQLite, Regex, Streamlit

GitHub | Live Demo

- Built a data pipeline cleaning and normalizing **254,000+** medicine records, engineering a canonical salt-matching key (regex-based composition parsing + alphabetical field sorting) to correctly group differently-branded drugs with identical composition
- Fixed a unit-price normalization bug for injectable dosage forms (per-volume vs. flat-dose strengths), preventing unfair price comparisons across differing pack sizes
- Wrote SQL analytics (GROUP BY, self-joins) over 6,235 comparable drug groups in SQLite, surfacing a **184% median price gap** between cheapest and costliest brand for the same salt; shipped both a static client-side site and a SQL-backed Streamlit app

Skin Disease Detection using CNN

Feb 2025

Python, TensorFlow, Keras, CNN, Computer Vision

GitHub

- Developed a CNN-based image classification model using TensorFlow/Keras to detect 7 categories of skin diseases, including melanoma, eczema, and psoriasis, from dermoscopic images
- Trained and validated the model on the HAM10000 dataset, applying data augmentation techniques to improve generalization across lesion types
- Addressed severe class imbalance in the dataset using targeted resampling and augmentation strategies, improving minority-class recognition, and achieved **90%+ test accuracy** through iterative tuning

Achievements

- **Top 25 Performer**, NeoCodeathon 2026 – Recognized for coding and problem-solving performance
- **Rank 29/250+**, AMD Ryzen Slingshot Regional Ideathon – VIT Bhopal University
- **Health Hack 2024** (VIT Bhopal – JHU) – Built an IoT-based Remote Health Monitoring System

Certifications

- Applied Machine Learning in Python – Coursera, Feb 2026
- Data Analysis with Python – IBM SkillsBuild, Sep 2025
- AWS Certification Course – Intellipaat, Sep 2025